

The Min-Max Loop

DATA 443/643 · Week 11, note 11.2

Week 11

American University

Ahmad Mousavi

Ideas from Stanford CS231N 2025 L13 (minimax; alternate D then G). Original slides; that course is not copied.

Lecture plan

- 1 Write J. Circle that G is only in the fake term.
- 2 One D step, one G step
- 3 Non-saturating $-\log D(G(z))$
- 4 A falling G loss is not better pictures

Key goal: a 99% D is often a dead G.

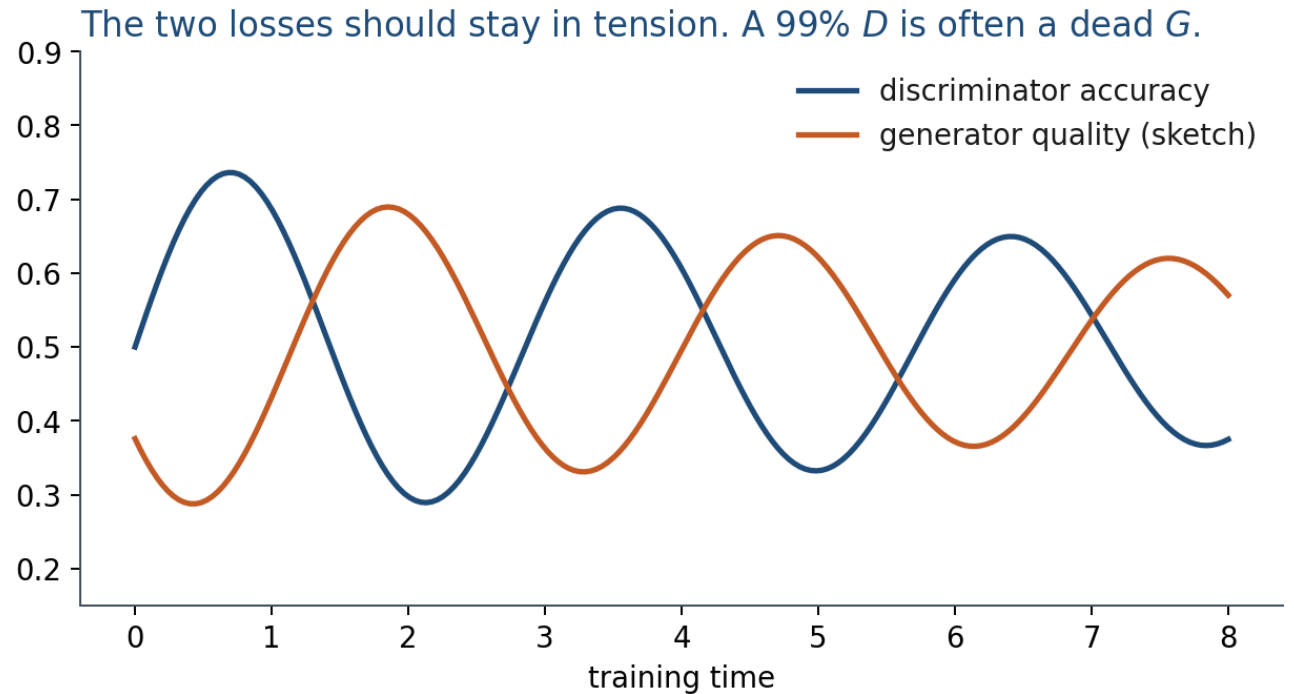
First time: what / why / formula

- What / why. D climbs to tell real from fake. G climbs to fool D . The original saturating loss for G is weak; people use $-\log D(G(z))$. If you only train D to 99%, G gets no gradient. If you only train G , D is stale.
- Architecture. Same two nets. Alternate (or simultaneous) SGD steps. No KL term here; that is RLHF.
- Formula. $J(D, G) = \mathbb{E}_x[\log D(x)] + \mathbb{E}_z[\log(1 - D(G(z)))]$. G minimizes a surrogate, not always this J .
- Tradeoffs. + When balanced, samples improve. – Oscillation, vanishing gradients for G , sensitivity to step sizes; next note is collapse.

How. Phase D: maximize $\log D$ on reals + $\log(1 - D)$ on fakes. Phase G: non-saturating $-\log D(G(z))$.

Keep the two losses in tension

- Train D on reals=1 and fakes=0. Freeze G.
- Train G with labels=1. Freeze D. You want D wrong.
- Detach fakes on the D step. One shared backward is a bug.



Who wants J large

$$J = \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$$

D maximizes J. G minimizes J.

G does not appear in the real term.

Nash hope: $p_g = p_r$ and
 $D = 1 / 2$. Training does not
promise it.

One real, one fake

- $D(x) = 0.9, D(G(z)) = 0.2$:
 $J \approx -0.328$.
- At $D(G) = 0.01$, $\log(1 - D) \approx -0.010$ is flat.
- $-\log D \approx 4.605$ still has slope. Lab 11 feeds ones into BCE.

$D(G(z)) = 0.01$: saturating G loss is flat; $-\log D$ still has slope.

quantity	value
J sample ($D_x = 0.9, D_g = 0.2$)	-0.328
$\log(1 - D)$ at 0.01	-0.010
$-\log D$ at 0.01	4.605

Two generator losses

Saturating

Minimize $\log(1 - D(G(z)))$. Early on, D is sharp and the slope dies.

Non-saturating

Maximize $\log D(G(z))$. Same game, usable gradient. That is the lab default.

What to take from this lecture

Two phases

Only θ_d , then only θ_g . Not one joint backward.

Do not rank

A small G loss can mean D collapsed.

Lab 11

400 steps. Print both losses. Then look at the scatter.